

Сетевые технологии

Лабораторная работа № 8.

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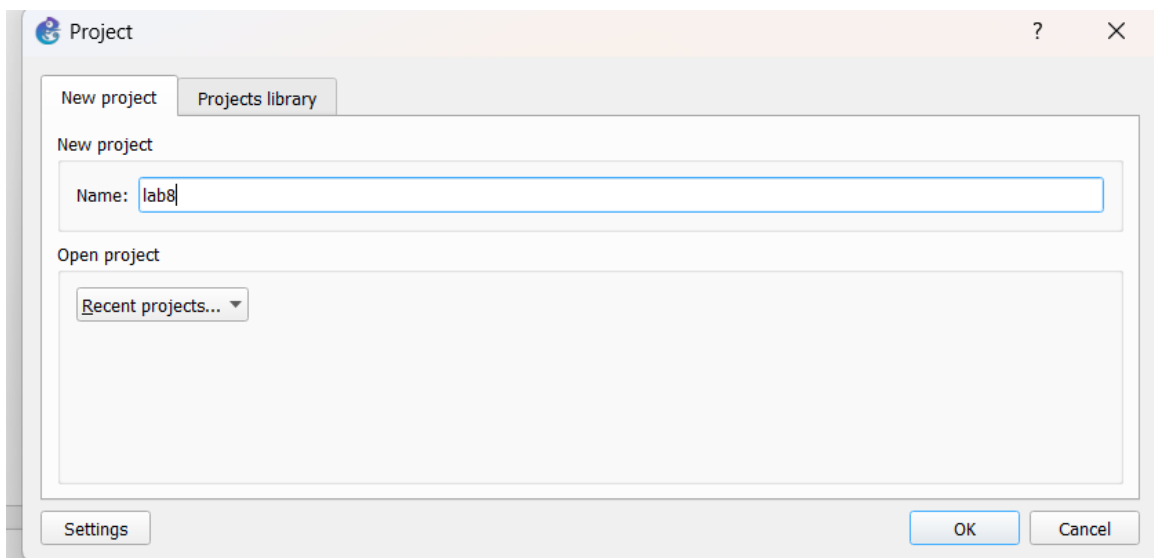
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1 Цель работы

IPv4- IPv6-

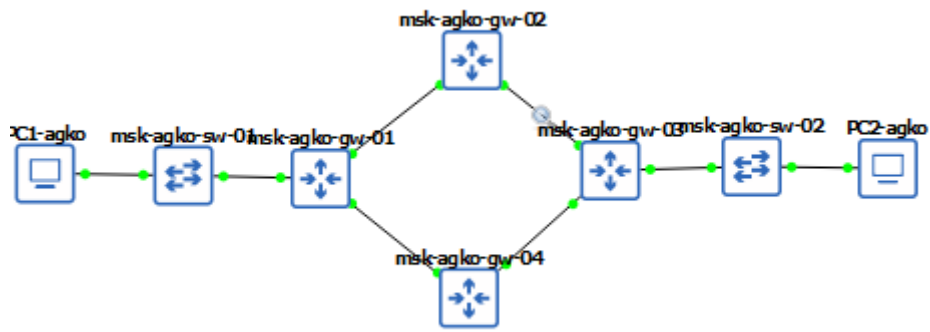
2 Выполнение работы

GNS3 «lab8» (. [#fig:001]).



2.1: GNS3

, , FRR (. [#fig:002]).



2.2: GNS3

IP- PC1 (. [#fig:003]).

```
PC1-agko> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
PC1-agko : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

PC1-agko> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
PC1-agko : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

PC1-agko> save
Saving startup configuration to startup.vpc
. done

PC1-agko> show ip

NAME          : PC1-agko[1]
IP/MASK       : 10.0.11.10/24
GATEWAY       : 10.0.11.1
DNS           :
MAC           : 00:50:79:66:68:00
LPORT        : 10016
RHOST:PORT    : 127.0.0.1:10017
MTU           : 1500

PC1-agko> 
```

2.3: IPv4 PC1

IP- PC2 (. [#fig:004]).

```
Executing the startup file

PC2-agko> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
PC2-agko : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

PC2-agko> save
Saving startup configuration to startup.vpc
. done

PC2-agko> show ip

NAME          : PC2-agko[1]
IP/MASK       : 10.0.11.10/24
GATEWAY       : 10.0.11.1
DNS           :
MAC           : 00:50:79:66:68:01
LPORT        : 10066
RHOST:PORT    : 127.0.0.1:10067
MTU           : 1500

PC2-agko> █
```

2.4: IPv4 PC2

gw-01 IPv4- eth0, eth1 eth2 (.
[#fig:005]).

```
msk-agko-gw-01 - PuTTY
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-user-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msk-user-gw-01#
```

2.5: gw-01

gw-02 IPv4- eth0 eth1 (. [#fig:006]).

```

Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-agko-gw-02# show running-config
Building configuration...

!
Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
exit
!
interface eth1
 ip address 10.0.2.1/24
exit
!
end
msk-agko-gw-02# █

```

2.6: gw-02

gw-03 IPv4- eth0, eth1 eth2 (.

[#fig:007]).

```
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
end
msk-agko-gw-03#
```

2.7: gw-03

gw-04 IPv4- eth0 eth1 (. [#fig:008]).

```
[OK]
msk-agko-gw-04# show running config
% Unknown command: show running config
msk-agko-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-agko-gw-04#
```

2.8:

gw-04

IPv6- PC1 (. [#fig:009]).

```
PC1-agko - PuTTY

PC1-agko> ip 2001:10:a/64
Invalid ipv6 address.

PC1-agko> ip 2001:10::a/64
PC1 : 2001:10::a/64

PC1-agko> save
Saving startup configuration to startup.vpc
. done

PC1-agko> show ipv6

NAME                : PC1-agko[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 2001:10::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPORT               : 10016
RHOST:PORT           : 127.0.0.1:10017
MTU:                 : 1500

PC1-agko> █
```

2.9: IPv6 PC1

IPv6- PC2 (. [#fig:010]).

```
PC2-agko - PuTTY

Checking for duplicate address...
PC2-agko : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

PC2-agko> ip 2001:11::a/64
PC1 : 2001:11::a/64

PC2-agko> save
Saving startup configuration to startup.vpc
. done

PC2-agko> show ipv6

NAME                : PC2-agko[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:11::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 10066
RHOST:PORT          : 127.0.0.1:10067
MTU                 : 1500

PC2-agko> 
```

2.10: IPv6 PC2

gw-01 IPv6 (forwarding) IPv6- (.
[#fig:011]).

```
msk-agko-gw-01 - PuTTY
Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-01
service integrated-vtysh-config
!
interface eth0
  ipv6 address 2001:10::/64
  ipv6 address 2001:10::1/64
  no ipv6 nd suppress-ra
exit
!
interface eth1
  ipv6 address 2001:1::1/64
exit
!
interface eth2
  ipv6 address 2001:4::2/64
exit
!
end
msk-agko-gw-01#
```

2.11: IPv6 gw-01

gw-02 IPv6 (. [#fig:012]).

```

msk-agko-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-agko-gw-02
service integrated-vtysh-config
!
interface eth0
  ipv6 address 2001:1::2/24
exit
!
interface eth1
  ipv6 address 2001:2::1/24
exit
!
interface eth-
exit
!
end
msk-agko-gw-02#

```

2.12: IPv6 gw-02

. RIP, OSPF . . . VyOS
 , 7 . . , .